Reliable Team Assistant: Phase 1

Yiting Dai | Solo | CSE 291 A Systems, Option 5 | October 24, 2026

Question and implemented scope

Can dependency-aware recovery preserve evidence-backed diagnosis quality with less recomputation after a trusted invalidation event? Configuration and telemetry roles collect read-only snapshots; a coordinator diagnoses. The runtime records consumed versions, invalidates dependent work, retains independent artifacts, rejects obsolete publications and links corrections. Missing recorded provenance triggers restart in development controls; this fallback was not exercised in the formal batch. Arbitrary error discovery, operational writes and invisible dependencies are outside the implemented guarantee.

Workload and frozen protocol

Cloud-OpsBench is pinned to ea05dafbc87b5985df8721622d63d4229c2b4224. Formal cases boutique/runtime/32, service/12 and performance/5 each run twice, reversing restart/selective order on repeat 2. These are three independent cases, not six. TritonAI api-glm-5.3 uses low reasoning, temperature 0 and 8192 output tokens. Course DataHub CPU 2/8 GB runs isolated Python 3.11.9 with exact dependency pins. Source 1380c6b identifies execution; the complete freeze is archived.

The user advanced launch to October 24 before formal observations were inspected. Cases, order, model, budgets and scoring were unchanged. Course acceptance of the explicitly user-selected benchmark remains unverified. Earlier development experiments and the 17-call TritonAI seen-case workflow remain separate feasibility evidence. No email or assignment submission was sent.

Formal quality results

All 6 units completed. Strict pass counts over all 6 assigned units: clean 4/6; restart 3/6; selective 3/6; prevention 3/6; expiry-restart 2/6; expiry-selective 2/6; no-recovery 0/6. For numeric-port recovery, only 4 units were eligible: restart and selective each passed 3/4, with different failed units. The other 2 units remain not-injected, not quietly excluded. Collection is not a diagnosis endpoint. No-recovery fails the current-publication criterion by construction; 0/6 does not imply six incorrect diagnoses. Prevention checks source integrity, not arbitrary semantic truth.

Strict success requires diagnosed status, native joint root/target accuracy and evidence closure, input-supported reasoning, current publication, valid output contract and budget. Input-only judgments on all 36 produced answers were hash-sealed before native labels were accessed. Codex performed the review; it is not independent human validation. All 54 assigned arm slots remain in scoring. Provisional publication does not mean operational approval.

Development, costs and limits

Formal result and measurement boundary

Among four eligible formal units, selective recovery used 1 new call versus 3 for restart: 33,260 versus 64,902 incremental tokens (48.8% fewer). Equal 3/4 quality counts had different failed cases, so do not establish quality equivalence. None of four corrupted ports propagated. Both methods needed 3 calls on all six all-expired units. Formal timing sums API-call latency; full workflow latency and integrated tracking overhead were not recorded.

Separate repeated handoff development: one seen case

handoff-dev-v2 ran three fresh prefixes on already-seen boutique/runtime/1 with fixed R/S, S/R, R/S recovery order. Specialists hand off summaries and a source catalog; the coordinator can retrieve 1-3 unchanged raw observations in one round. A fixed summary edit changes httpGet to grpc. The raw-visible baseline has different evidence exposure and call budget. All 21 prescribed arm slots are retained, including three invalid no-recovery arms; none were replaced.

Paired recovery work, not total service savings

Each repeat used 6 restart calls versus 3 selective calls. Restart/selective incremental tokens: 24,553/12,163; 25,825/12,176; 25,660/12,020 (50.5%, 52.9%, 53.2% fewer). Incremental recovery-arm wall seconds: 14.319/6.640; 15.665/6.813; 14.459/6.169. These timers exclude notification, invalidation, cloning and final evidence persistence. Means were 25,346/12,119.7 tokens and 14.814/6.541 seconds. Selective retained eight byte-identical valid artifacts per repeat; provenance and freshness checks alone do not establish semantic correctness.

Semantic evidence

Independent AI review was sealed before method mapping: all 15 core diagnoses were source-supported; corrupted handoff and raw-visible baselines each adopted 0/3 false grpc claims. Paired review found both methods corrected the intermediate summary in 3/3 repeats and restored six version-valid publications, but prior final diagnoses were already correct. No-recovery rejected stale publication, not a wrong diagnosis. Probe-timing errors occurred in clean repeat 1 and selective repeats 1 and 3, with none in restart; ancillary claims remain unverified. The reviewer had prior case context and is not human. Thus wrong-final-diagnosis repair and whole-answer equivalence remain unproven. One seen case repeated three times supports no significance, p95 or generalization claim.

Separate costs and next decision

The immutable formal batch used 92 calls, 519,696 tokens, $0.40225095 in reconciled header cost, and no API errors. The new handoff batch separately used 54 calls, 241,475 tokens and $0.2136544; its settled course balance was $49.11201275. Shared-prefix costs are excluded from paired incremental comparisons. The earlier single-run causal-summary experiment remains separate in the archive. Develop a propagation-producing protocol before freezing a new evaluation; exposed cases cannot become unseen tests.

Reproduction and timeline appendix

Portable evidence and immutable formal archive

The ZIP contains the pinned Cloud-OpsBench evaluator/labels/license, original formal traces and hash seal, all handoff raw requests/responses, costs, snapshots, dependency and reuse records, blinded review, runner, verifier and lockfile. Original formal evidence is byte-identical. Clean extraction supports saved-output verification without credentials or inference; package installation can require internet.

Offline verification after extraction

Create a Python >=3.11 environment and install requirements.lock. Run python scripts/verify_formal_release.py from the extracted root. It disables network access, checks release hashes, the original review seal, all 36 produced formal answers and all 54 assigned formal slots. Run python scripts/summarize_handoff_saved.py development/handoff-dev-v2 to recompute handoff usage, paired costs and reuse checks. Saved-output checks do not reproduce mutable model responses or replace independent semantic review. See docs/phase1-handoff-reproduction-20261024.txt for exact commands and the blinded-review/packet seal check.

Completed October 24

Frozen formal evaluation, native scoring, sealed evidence judgments, billing reconciliation and portable reproduction. Separate handoff development: three repeats on one seen case, fixed fault and order, all outcomes retained. AI assistance includes implementation, analysis and drafting. Independent AI review is distinct from student/human validation. No submission or operational action was performed.

October 24-November 4

Complete student review and extraction checks; preserve nonpropagation and unsupported claims. Refine the causal-fault protocol only as a new development version. The saved course overview lists November 4 as the updated Phase 1 target; verify the exact live deadline and upload slots before any submission. Benchmark course acceptance remains unverified.

Next experimental milestone

Only after development supports propagation, freeze fault eligibility, matched failure states, recovery quality and cost criteria; include complete invalidation and missing provenance. Select independent evaluation cases excluding all exposed cases. Recheck key/model expiry, quota and explicit budget before any online batch; this archive does not authorize inference.

Remaining term and limitations

Analyze paired quality before resource savings. Multi-group scheduling and gated procedural learning follow the core recovery evidence. Prepare final artifacts/report/presentation in late November-December, checking official dates. Static snapshots, trusted invalidation and mutable model aliases limit claims. Sources: Cloud-OpsBench repository; saved course Option 5 overview and Piazza 46; TritonAI documentation.